

Fuel Pump Queue Management System

Code : <https://www.programiz.com/online-compiler/6fAwvCtzXvgz8>

1. Problem Statement

At a fuel pump, many vehicles may arrive at the same time to get fuel. Managing these vehicles manually can create long queues, confusion, and unnecessary waiting time. It can also be difficult to maintain the correct order of vehicles.

2. Proposed Solution

The Fuel Pump Queue Management System provides a simple solution using a **Queue data structure**. Vehicles are added to the queue when they arrive at the fuel pump. The vehicle at the front of the queue is served first. After receiving fuel, that vehicle leaves the queue and the next vehicle is served.

The system follows the **FIFO (First In, First Out)** principle, which means the vehicle that arrives first gets fuel first.

3. Used Data Structure

Queue:

Used to maintain the waiting vehicles. New vehicles are added at the **rear**, and vehicles are served from the **front**.

Array:

Used to store the vehicle numbers in the queue.

FIFO:

The system follows **First In, First Out**, which is suitable for managing a real-life fuel pump queue.

6. Benefits

- Reduces confusion at the fuel pump.
- Maintains the correct order of vehicles.
- Demonstrates the real-life use of a Queue.
- Easy to understand and implement.
- Follows the FIFO principle.

Output:

```
1.Add Car  2.Give Fuel  3.Display  4.Exit  
Enter choice: 1  
Enter Car Number: 101  
Car added.
```

```
1.Add Car  2.Give Fuel  3.Display  4.Exit  
Enter choice: 2  
Car 101 got fuel.
```

```
1.Add Car  2.Give Fuel  3.Display  4.Exit  
Enter choice: 3  
Queue Empty!
```

```
1.Add Car  2.Give Fuel  3.Display  4.Exit  
Enter choice: 4
```

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